

Definability of quantization inputs: a Padoa-style classification and its boundary

Igor Postanovsky*

August 3, 2026

Abstract

Quantum theory carries several discrete or numerical inputs that are routinely described as “not derived” from an underlying substrate. We ask a sharper question: for which of these inputs can “not derived” be upgraded to a proof of undefinability, and for which is this structurally impossible? Padoa’s classical method answers the first kind and, we show, cleanly refuses the second. We propose a classification of quantization inputs by their logical position: an input is amenable to Padoa’s method exactly when it is a predicate on a fixed similarity type, and is not amenable when it is a parameter of the type itself or a weight on an ensemble. We then exhibit both poles. At one pole, circulation quantization: reformulating a companion result, we show that the predicate is Padoa-undefinable over the structure of realized jets, with a pair whose existence is the nontrivial part; and we observe that here the method yields more than a verdict, since it computes the exact reduct threshold at which definitions begin to exist, direction by direction. At the other pole, the spinor sign: the pair is nearly definitional, and the entire mathematical weight sits in calibrating the reduct, which we isolate as an explicit hypothesis rather than assert. Finally we show that the amplitude parameter and Born-type typicality weights fall outside the method. For the amplitude parameter we answer the natural objection that deformation quantization supplies a common type, and prove a dichotomy for the single-theory formulation: with the external potential in the reduct the parameter is definable away from a degenerate stratum, and with it removed the parameter is undefinable only because a free function absorbs the difference. For typicality weights we document a semantic obstruction with a counterexample. In both cases the two available branches give, respectively, an answer without blindness and blindness without content, which is our reason for excluding these inputs structurally. The boundary is offered as a result, not an apology: the classification tells one in advance which foundational “inputs” can be argued about by this method at all.

1 Introduction

Every axiomatization of quantum mechanics arrives with a short list of ingredients that the formalism uses but does not produce. The value of the reduced Planck constant sets the scale of noncommutativity; the circulation of a velocity field is quantized in integer multiples of it; a spinor changes sign under a full rotation while a vector does not; the Born rule weights configurations by a squared amplitude. Foundational writing tends to mark each of these with the same phrase — “put in by hand”, “postulated”, “not derived” — and to pass on. This paper takes that phrase seriously enough to ask, input by input, when it can be turned into a theorem and when it cannot.

*Independent researcher, Germany. ORCID: 0009-0000-5338-7809. This preprint: Zenodo DOI 10.5281/zenodo.21769912. Supplementary code: Zenodo DOI 10.5281/zenodo.21769869.

The question. Foundational discussions routinely record that a certain ingredient of quantum theory is “put in by hand”. Such statements are claims about the absence of a derivation, and absence of a derivation is not a theorem. Padoa’s method converts some of them into theorems: a notion P is undefinable from primitives Σ within a theory T if two models of T agree on all of Σ and differ on P . Our question is which quantization inputs admit this treatment.

Related work. Applying Padoa’s method to physical theories is an established programme rather than a new move, and we position this paper inside it. Padoa’s principle [2] has been applied to the McKinsey–Sugar–Suppes axiomatization of classical particle mechanics [5] to discuss the definability of physical notions such as mass and time [7], with Tarski [4] and Beth [3] supplying the theoretical grounding of the method and Suppes’ textbook formulation [6] the version most often quoted in that literature. The same programme has been continued by Sant’Anna and Bueno [8], alongside earlier work on the definability of time and spacetime in classical physics [9]. What appears to be untried is the application to *quantization conditions* specifically, and that is the slot this paper occupies.

We must also distinguish a different use of the word. There is a line of work, due to BenDaniel [10], in which “definability of physical fields” means set-theoretic definability within an axiomatic set theory, and in which quantization of fields is claimed to be equivalent to definability in that theory. Despite the superficial closeness of the vocabulary, that programme asks whether field configurations are definable *sets*, whereas we ask whether a *predicate* is definable from a *declared reduct* of physical primitives. The two notions of definability are not comparable, and no result of either transfers to the other. A further distinction separates our sense from a recent application of Tarski’s undefinability theorem to physics [13], which concerns the undefinability of truth within a formal language rather than the definability of a designated predicate from a declared reduct; the two share the word and little else.

Contribution. (i) A classification of quantization inputs by logical position (Section 3), which we regard as the main content; (ii) the two poles instantiated (Sections 4 and 5); (iii) the boundary argued rather than asserted (Section 6).

2 Preliminaries: what a Padoa statement needs

Let T be a theory in a similarity type containing primitives Σ together with a distinguished symbol P .

Definition 2.1 (Padoa pair). A *Padoa pair for P over Σ in T* is a pair of models M, M' of T whose Σ -reducts coincide, $M \upharpoonright \Sigma = M' \upharpoonright \Sigma$, and for which P receives different interpretations.

If such a pair exists, P is not explicitly definable from Σ in T ; by Beth’s theorem the converse holds in first-order logic. In this paper only the easy direction is used — a pair excludes an explicit definition — and that direction is valid in any semantics, including the jet-image semantics of Section 4.

Two calibrations, not one. A Padoa statement is calibrated by *two* declarations, and omitting either makes it vacuous or unintelligible.

- The **reduct** Σ . Enlarging Σ strengthens an undefinability claim: it becomes harder for a pair to exist. Shrinking Σ weakens it. A claim quoted without its reduct has no content.

- The **theory** T . Both members of the pair must be models of *the same* T . If T already constrains P — for instance by an axiom fixing its value — no pair exists and the question is closed by fiat. Conversely, a T chosen so loosely that it says nothing about P makes undefinability trivial. The interesting statements are those where T is the natural theory of the domain and P is left open by it, and this is exactly what has to be argued in each case.

We therefore state T and Σ explicitly before each pair, and we say in each case where the burden of the theorem lies.

3 The classification

We propose that quantization inputs divide by their *logical position* relative to a similarity type, and that this position — not their physical importance or their discreteness — decides whether Padoa’s method can address them at all.

Position of the input	Padoa’s method	Instance
predicate on a fixed type, pair hard to construct	applies; content is the pair	circulation quantization (Sec. 4)
predicate on a fixed type, pair immediate	applies; content is the calibration of Σ	spinor sign (Sec. 5)
parameter of the type itself	does not apply; dichotomy proved (Prop. 6.1)	amplitude parameter (Sec. 6.1)
weight on an ensemble	does not apply; semantic obstruction (Sec. 6.2)	Born-type typicality (Sec. 6.2)

The first two rows share a type and differ in where the work sits; the last two are not failures of ingenuity but structural exclusions, and Section 6 argues this rather than assuming it. The practical value of the classification is predictive: given a new “input”, one can ask which row it occupies before spending effort on a proof that cannot exist. We do not claim the four positions exhaust the possibilities: the classification is a working taxonomy, offered for its predictive use rather than as a proven partition, and an input fitting none of the four rows would be a finding rather than a counterexample to anything asserted here.

4 First pole: circulation quantization (hard pair, natural reduct)

4.1 Theory and reduct

T_{hydro} . Models are hydrodynamic data $d = (\rho, v)$ on $Q = S^1$: ρ a smooth positive function with

$$\int_{S^1} \rho d\theta = 1, \quad (1)$$

and v a smooth 1-form, necessarily closed on S^1 . Write $v = g d\theta$ and

$$\text{per}(d) = \oint_{S^1} g d\theta. \quad (2)$$

T_{hydro} imposes no condition on per ; this is the physically standard class and the point of departure of the discussion initiated by Wallstrom [11].

Σ_{jet} . The primitives are the realized jets: for each k , the set

$$J_k(d) = \{ j_\theta^k(\rho, v) : \theta \in Q \} \quad (3)$$

of k -jets of (ρ, v) at points of Q , with arbitrary predicates on finite tuples of jets and quantifiers ranging over points, with *jet-image semantics*: the truth value of any Σ_{jet} -statement on d is a function of the family $(J_k(d))_k$.

P . The circulation-quantization predicate: $\text{per}(d) \in 2\pi\mathbb{Z}$, in units with the reduced Planck constant set to one.

4.2 The pair, self-contained

Set $\rho \equiv 1/(2\pi)$, so the density contributes one constant jet and plays no further role. Let $s : [0, 1] \rightarrow [0, 1]$ be the standard smooth step,

$$s(x) = \frac{f(x)}{f(x) + f(1-x)}, \quad f(x) = \begin{cases} e^{-1/x}, & x > 0, \\ 0, & x \leq 0; \end{cases} \quad (4)$$

all derivatives of s vanish at both endpoints, and $s(x) + s(1-x) = 1$, whence $\int_0^1 s = 1/2$.

Fix levels $a_1 \neq a_2$ and a width $\tau > 0$. For $L_1, L_2 > 0$ with $L_1 + L_2 + 2\tau = 2\pi$ define $g = g_{L_1}$ piecewise:

$$g(\theta) = \begin{cases} a_1, & \theta \in [0, L_1), \\ a_1 + (a_2 - a_1) s\left(\frac{\theta - L_1}{\tau}\right), & \theta \in [L_1, L_1 + \tau), \\ a_2, & \theta \in [L_1 + \tau, L_1 + \tau + L_2), \\ a_2 + (a_1 - a_2) s\left(\frac{\theta - L_1 - \tau - L_2}{\tau}\right), & \theta \in [L_1 + \tau + L_2, 2\pi). \end{cases} \quad (5)$$

Since every derivative of s vanishes at both endpoints, g is smooth on S^1 , and $(\rho, g d\theta)$ is a model of T_{hydro} .

Jet image. For any k , the k -jet at a point depends only on which of the four pieces the point lies in and on its local coordinate within that piece. On the plateaus it is the constant jet $(a_1, 0, \dots, 0)$, respectively $(a_2, 0, \dots, 0)$; on the transitions it is the jet of the fixed profile at local coordinate $u \in [0, 1]$, a function of u, τ, a_1, a_2 alone. The four pieces are the same for all admissible (L_1, L_2) — only the plateau lengths and the placement change. Hence

$$J_k = \{(a_1, 0, \dots, 0)\} \cup \{(a_2, 0, \dots, 0)\} \cup J_k(\text{descent}) \cup J_k(\text{ascent}) \quad (6)$$

is independent of (L_1, L_2) , for every k . This is an identity of sets established structurally; no limiting or numerical argument enters.

Circulation. Integrating piecewise, the two transitions contribute $\tau(a_1 + a_2)$ in total, so

$$\text{per}(g_{L_1}) = a_1 L_1 + a_2 L_2 + \tau(a_1 + a_2), \quad L_2 = 2\pi - 2\tau - L_1, \quad (7)$$

affine in L_1 with nonzero slope $a_1 - a_2$.

A pair. Take $a_1 = 2$, $a_2 = 0$, $\tau = 1/5$, so $\text{per} = 2L_1 + 2/5$. Then

$$L_1 = \pi - \frac{1}{5} \implies \text{per} = 2\pi, \quad L_1 = \pi + \frac{3}{10} \implies \text{per} = 2\pi + 1. \quad (8)$$

Both leave L_1 and L_2 positive, so both are models of T_{hydro} ; by (6) their jet images agree in every order.

Proposition 4.1 (restatement of [1, Thm. 3.7] in Padoa form). *The circulation-quantization predicate is Padoa-undefinable over Σ_{jet} in T_{hydro} .*

Proof. The two data above have equal Σ_{jet} -reducts — equal, not merely elementarily equivalent — and differ on P . $\square$

Remark 4.2 (where the burden sits). Here the reduct is the natural one and the theory is the standard class; the entire difficulty is the *existence* of the pair, because jets must come from an actual function on the circle. That rigidity is what the plateau surgery defeats. Note also that the boundary of Σ_{jet} is computed rather than chosen: germ-type structures are deliberately excluded, and by [1, Cor. 4.2] they read exactly the period modulo an explicitly computed subspace.

Remark 4.3 (what is and is not new here). Proposition 4.1 is a translation of a companion result into the language of Section 2; we claim no new mathematics for it. It is included because the classification of Section 3 needs an instantiated first row, and because the translation is what connects the companion to the tradition cited in Section 1.

Corollary 4.4 (the definability threshold). *Proposition 4.1 together with [1, Thm. 4.1 and Cor. 4.2] does more than exclude: it locates the reduct at which definability of the circulation predicate switches on, and does so direction by direction. Over Σ_{jet} the predicate is Padoa-undefinable. Over the germ-along- K reduct, each direction of the circulation datum is either invisible to K or functionally determined by the coset test — the period vector modulo*

$$V_K = \text{im}(H_c^1(Q \setminus K) \longrightarrow H^1(Q)) \quad (9)$$

— and there is no intermediate regime. Definability therefore switches on with the cycle coverage of K , one direction at a time.

Two features of this are worth isolating. First, it makes the first row of the classification *quantitative*: for this input the method does not merely report undefinability over a chosen reduct but computes the exact reduct threshold at which definitions begin to exist. Second, the threshold is a genuine dichotomy rather than a gradual loss of information — a direction is read exactly or not at all — which is what allows it to be stated as a threshold in the first place. We are not aware of an analogue in the Padoa-in-physics literature cited in Section 1, where undefinability claims are made relative to one fixed reduct.

5 Second pole: the spinor sign (immediate pair, calibrated reduct)

5.1 Theory and reduct

T_{frame} . Models are pairs $M = (R, \chi)$ where $R: [0, 1] \rightarrow SO(3)$ is a smooth closed path — the rotation history of a frame carried along a worldline — and χ is a character of $\pi_1(SO(3)) = \mathbb{Z}_2$, that is, a homomorphism into $\{+1, -1\}$. T_{frame} imposes no condition on χ . That it does not is the substantive point, argued in Section 5.3: the theory of framed worldline geometry does not decide the character.

Σ_{spin} .

- (i) the path $R(t)$ and all its functionals;
- (ii) all integer-spin local observables;
- (iii) the infinitesimal data $\omega(t) = R'(t)R(t)^{-1} \in \mathfrak{so}(3)$;
- (iv) the evaluations of χ on contractible loops.

P. The predicate that χ is the nontrivial (sign) character — equivalently that the frame transported around the noncontractible 2π rotation returns with a sign.

5.2 The pair

Theorem 5.1. *P is Padoa-undefinable over Σ_{spin} in T_{frame} .*

Proof. Let $M_+ = (R, \chi_+)$ with χ_+ trivial and $M_- = (R, \chi_-)$ with χ_- the sign character, for one and the same R . Both are models of T_{frame} . For (i): the reducts share R literally — the two $SU(2)$ lifts $\pm U$ of a rotation project to the same element of $SO(3)$. For (ii): integer-spin representations of $SU(2)$ factor through $SO(3)$, hence every integer-spin observable is a function of $R(t)$ and takes equal values in both models. For (iii): ω is computed from R alone. For (iv): a character sends the identity class to 1, so both characters agree on contractible loops. Hence $M_+ \upharpoonright \Sigma_{\text{spin}} = M_- \upharpoonright \Sigma_{\text{spin}}$. On the noncontractible class the interpretations differ, +1 against -1. $\square$

Remark 5.2 (why including χ on contractible loops is not circular). Item (iv) mentions the target symbol, and a reader may suspect that a fragment of P has been smuggled into the reduct. It has not, and the point is worth stating rather than leaving to inference. The evaluations in (iv) are constant across *all* models of T_{frame} — every character is 1 on the identity class — so (iv) carries no information whatever about which character is present. It is included precisely because including it costs nothing and makes the reduct larger, and a larger reduct yields a strictly stronger undefinability claim: the sign remains undefinable even when the character’s values on contractible loops are given.

5.3 The calibration, stated as a hypothesis rather than asserted

Unlike Section 4, the pair here is immediate: characters factor through π_1 by definition, so two characters on the same base path always exist. The mathematical content is therefore entirely in the claim that Σ_{spin} is *physically complete* — that nothing local has been left out. We isolate it.

Definition D (local geometric observable). A *local geometric observable* of a framed worldline is a functional whose value at each parameter t is determined by a finite jet of R at t — a smooth function of $R(t)$ and finitely many derivatives, equivalently of $R(t)$ together with the angular-velocity data $\omega(t) = R'(t)R(t)^{-1}, \omega'(t), \dots$ — transforming tensorially under a global frame rotation.

Definition D is a stipulation about the scope of the word “local”, and the entire weight of Theorem 5.1 rests on accepting it as the right stipulation for a framed worldline. We therefore state the assumption separately, in the form in which it is contestable.

Hypothesis L (locality scope). A local measurement performed on a framed worldline is confined to local geometric observables in the sense of Definition D; in particular no such measurement has access to the value of the character χ on a noncontractible loop.

Lemma 5.3. *Every local geometric observable in the sense of Definition D takes equal values on M_+ and M_- , so none defines P .*

Proof. The two models share R literally, hence share every jet of R and every derivative of $\omega = R'R^{-1}$; any functional built from those data takes equal values in both. The sign character is not a functional of that kind: it is a homomorphism on $\pi_1(SO(3))$ evaluated on loop classes, and the pair fixes one and the same R — hence the same loops — while assigning the noncontractible class opposite character values. No finite jet of R at any point carries the value of a homomorphism on a global homotopy class. $\square$

The Lemma is not itself the content: given Definition D it is immediate, since the two models share R literally. Its function is to show that Hypothesis L is the *only* thing assumed — once the scope of “local” is fixed, no further representation-theoretic input is required, and none is smuggled in. What remains contestable is therefore a single, clearly located claim about physics rather than a claim about algebra.

Illustration (the integer-spin sub-case of Definition D). Two facts calibrate the hypothesis. The projection identity behind (i) — that the two lifts $\pm U$ determine one and the same R — is verified symbolically to machine exactness in the Supplement. The closure behind (ii) is the classical Clebsch–Gordan fact that tensor products of integer-spin representations contain only integer spins; a definition of the sign from data in (i)–(iii) would have to manufacture a half-integer object from integer ones.

The residual, precisely. The Lemma does not settle Hypothesis L, and it is worth saying exactly what it leaves open. It does not prove that a local measurement is confined to jet-functionals of the frame; it proves that *if* it is, the sign is invisible. A reader may propose a broader notion of “local” — one that grants a local apparatus access to the character itself — under which L fails and the spinor sign becomes locally accessible. The disagreement is then not mathematical but concerns the scope of “local”, and Theorem 5.1 is the conditional statement that on the jet-functional reading of Definition D the sign is not recoverable from local geometry. A referee who rejects Definition D and Hypothesis L rejects the interest of Theorem 5.1 while leaving its proof intact — the correct division of labour, now located at the definition rather than left implicit.

Remark 5.4 (deflation, explicitly). Theorem 5.1 is close to the observation that $SU(2) \rightarrow SO(3)$ is a nontrivial double cover. We say so plainly. Its role in this paper is not to be difficult but to occupy the second row of the classification: an input for which Padoa’s method applies and returns an answer whose entire weight has migrated into the declaration of the reduct. The contrast with Section 4 — where the reduct is obvious and the pair is hard — is the point.

6 The boundary: inputs the method cannot reach

6.1 The amplitude parameter

Our claim is that the amplitude parameter is not a predicate on a fixed similarity type but a parameter of the type: it controls whether an oscillatory correlation structure is present at all.

The objection. Common types in which the parameter appears explicitly are standard. Deformation quantization presents a family of star-products indexed by the parameter; the Madelung system itself contains it as a coefficient in the quantum-potential term. Does this not supply exactly the setting Padoa needs?

Reply. It supplies a family, not a theory, and the distinction is what Padoa’s method turns on. Two structures with different values of the parameter are models of *different* members T_h of a family, not two models of one T ; Definition 2.1 requires the latter. One may of course build a single theory T in which the parameter is a constant symbol and the data are required to satisfy the corresponding equations. That formulation can be carried out, and it does not rescue the objection: it returns a dichotomy, neither branch of which is a contentful undefinability claim.

Proposition 6.1 (the single-theory formulation, and why it does not help). *Let T_{mad} have models (ρ, S, V, h) on the circle satisfying the continuity equation and the quantum Hamilton–Jacobi equation,*

$$\partial_t \rho + \partial_\theta \left(\frac{\rho \partial_\theta S}{m} \right) = 0, \quad (10)$$

$$\partial_t S + \frac{(\partial_\theta S)^2}{2m} + V - \frac{h^2}{2m} \frac{\partial_\theta^2 \sqrt{\rho}}{\sqrt{\rho}} = 0, \quad (11)$$

with $\rho > 0$ normalised and h a constant symbol. Then:

(i) With V in the reduct. *If two models of T_{mad} share (ρ, S, V) , subtracting the two copies of (11) leaves*

$$\frac{h_2^2 - h_1^2}{2m} \frac{\partial_\theta^2 \sqrt{\rho}}{\sqrt{\rho}} \equiv 0, \quad (12)$$

so either $h_1^2 = h_2^2$ or the quantum-potential term vanishes identically. Hence, fixing the sign convention $h > 0$, the parameter is implicitly — indeed explicitly, via the recovery expression

$$h^2 = 2m \left(\partial_t S + \frac{(\partial_\theta S)^2}{2m} + V \right) \frac{\sqrt{\rho}}{\partial_\theta^2 \sqrt{\rho}} \quad (13)$$

— definable over the subclass of models with nonuniform density. Padoa’s method here returns definability, not undefinability.

(ii) The degenerate stratum. *On the circle, $\partial_\theta^2 \sqrt{\rho} \equiv 0$ together with periodicity forces ρ constant, and normalisation then gives $\rho \equiv 1/(2\pi)$; equation (10) in turn forces $\partial_\theta^2 S \equiv 0$, so the velocity is constant. The stratum on which the parameter is invisible is therefore exactly the uniform-density, constant-velocity family — the same family singled out in the sharpness analysis of [1, Sec. 3.4(i)]. Undefinability does hold there, but on a stratum carrying no dynamical content.*

(iii) With V outside the reduct. *For any (ρ, S) and any value of h one may define V by solving (11) for it; the resulting triple is a model. Two different values of h then yield two models agreeing on (ρ, S) and differing on h , so the parameter is undefinable — trivially, because a free function has absorbed the difference.*

Verification. The identity (12), the two implications in (ii), and the constructions in (iii) were checked symbolically; the inverse problem in (i) was tested two-sidedly, recovering a preassigned value exactly rather than merely returning a nonzero one (Supplement, block C).

Remark 6.2 (the shape of the argument, and what it does not say). Neither branch of Proposition 6.1 is a contentful undefinability result, and the two branches have the same shape as the two semantics of Section 6.2: one gives an answer without blindness, the other blindness without content. This is why we place the amplitude parameter in the third row of Section 3 rather than the first.

It must be said plainly what (i) does *not* mean. Recovering the *numerical value* of the parameter inside T_{mad} is parameter identification within a theory that already contains the

quantum-potential term; it is not a derivation of quantum structure from classical data, and nothing here bears on whether that structure is derivable. The two claims are easy to conflate and are unrelated. Nor does this section engage the separate proposal to reformulate quantum mechanics without Planck’s constant as a dimensional primitive [12], which changes the status of the parameter rather than asking whether it is definable from a reduct; the two questions do not bear on one another.

Reopening criterion. Should a single theory T with a common type be exhibited in which the amplitude parameter is a predicate over a physically natural reduct, the question returns to the first row of Section 3 and this section becomes obsolete. We regard that as a desirable outcome, not a threat.

6.2 Typicality weights

Let E be a measure on a space of configurations, with support $\text{supp}(E)$. Statements about E come in two semantics, and neither yields a Padoa-style blindness claim with content.

Under *pointwise* semantics — quantifiers ranging over the points of $\text{supp}(E)$ — the truth value of any combination of per-configuration predicates is by construction a function of $\text{supp}(E)$. Blindness to the weights is then immediate and vacuous: it has been arranged by the choice of semantics.

Under *almost-everywhere* semantics the situation reverses: the weights are visible. Take

$$E_1 = \text{Leb} \upharpoonright [0, 1], \quad E_2 = \tfrac{1}{2}(\text{Leb} \upharpoonright [0, 1] + \delta_{1/2}). \quad (14)$$

Their supports are equal, both $[0, 1]$. The statement “for E -almost every x , $x \neq 1/2$ ” holds for E_1 and fails for E_2 , since $E_1(\{1/2\}) = 0$ while $E_2(\{1/2\}) = 1/2$.

Hence: one semantics gives blindness without content, the other gives content without blindness. A Padoa-style argument in this setting would need a third option, and we do not have one. This is our reason for placing typicality weights in the fourth row of Section 3.

Remark 6.3. The obstruction is specific to measures and does not arise in Section 4, where the analogous “support” — the jet image — is the exact image of a continuous map on a compact space, so that the realized set, its closure, and the support of the induced measure coincide. On an infinite-dimensional configuration space these three objects separate, which is precisely why the argument of Section 4 does not lift.

7 Discussion

The paper has sorted four quantization inputs by a single criterion — the logical position each occupies relative to a similarity type — and found that this position, not the physical weight of the input, decides whether Padoa’s method can reach it. Two inputs sit inside the method: circulation quantization, where the reduct is the obvious one and the whole difficulty is manufacturing a pair against the rigidity of jets (Section 4); and the spinor sign, where a pair is immediate and the whole content has migrated into a declaration of what counts as local (Section 5). Two inputs sit outside it: the amplitude parameter, a parameter of the type rather than a predicate on it, for which the single-theory formulation returns a dichotomy with no contentful branch (Proposition 6.1); and Born-type typicality weights, for which the two available semantics give blindness without content or content without blindness (Section 6.2). The same shape recurs at both excluded inputs, and it is that shape — an answer without blindness on one side, blindness without content on the other — that we read as the signature of a structural exclusion rather than a failure of effort.

At the first pole the method does more than deliver a verdict over one chosen reduct: by Corollary 4.4 it computes the threshold at which definitions begin to exist, direction by direction. That is the sense in which the first row of the classification is quantitative, and it is the feature we would most like to see tested against other inputs.

The classification is a working taxonomy, not a closed list, and its interest is that it extends. Other discrete ingredients of quantum theory — a topological angle, the integer labels of a superselection sector — present themselves as candidates for the same triage, and the criterion says in advance where to look: whether the ingredient is a predicate on a fixed type, so that a pair is worth hunting; a parameter of the type, so that a dichotomy is all one will find; or a weight, so that the measure-theoretic obstruction of Section 6.2 applies. Carrying the classification to those ingredients is left to future work.

Three remarks we would make regardless. First, the classification of Section 3 is falsifiable in a useful way: exhibiting a Padoa pair for either input of Section 6, over a declared reduct and a single theory, refutes the placement. Second, the boundary of Section 6 is argued on both sides rather than declared: for each excluded input we exhibit the two natural formulations and show what each returns. Third, the two theorems are not offered as difficult results — one is a translation and one is close to a textbook fact. They are offered as calibration points: they show what the two amenable rows look like from the inside, which is what makes the exclusion in Section 6 informative rather than merely negative.

Open questions. (a) Whether the dichotomy pattern of Proposition 6.1 — an answer without blindness on one branch, blindness without content on the other — can be promoted to a membership *criterion* for the third row is left open; any such promotion must first declare the class of single-theory formulations quantified over, on pain of exactly the semantic pitfall documented in Section 6.2. (b) *Speculative*: the stratum on which the amplitude parameter is invisible (Proposition 6.1(ii)) coincides with the constant-circulation family that carries the sharpness examples of [1, Sec. 3.4(i)]; whether “parameter-invisibility strata” and “sharpness families” coincide beyond this instance is not known to us.

Code availability

The symbolic checks reported in Sections 4, 5, 6.1 and 6.2 are collected in a self-contained SymPy script, `P2_Padoa_supplementary.py`, archived at Zenodo, DOI: 10.5281/zenodo.21769869. It runs 25 exact checks in four blocks, one per construction, exits with a nonzero status if any check fails, and depends only on SymPy. Every check is symbolic; none relies on floating-point sampling. The blocks include both halves of a two-sided test of Proposition 6.1: the inverse problem returns the preassigned value of the parameter uniquely off the degenerate stratum, and returns nothing on it.

References

- [1] I. Postanovsky, *Locality, cycles, and the Wallstrom condition: an exact dichotomy and an undefinability theorem*, Zenodo (2026). DOI: 10.5281/zenodo.21727985.
- [2] A. Padoa, *Essai d’une théorie algébrique des nombres entiers, précédé d’une introduction logique à une théorie déductive quelconque*, Bibliothèque du Congrès International de Philosophie **3** (1900) 309–365.
- [3] E. W. Beth, *On Padoa’s method in the theory of definition*, Indagationes Mathematicae **15** (1953) 330–339 [= Kon. Ned. Akad. Wet., Proc. Ser. A **56** (1953) 330–339].

- [4] A. Tarski, *Einige methodologische Untersuchungen über die Definierbarkeit der Begriffe*, Erkenntnis **5** (1935) 80–100; English transl.: *Some Methodological Investigations on the Definability of Concepts*, in *Logic, Semantics, Metamathematics*, 2nd ed., Hackett, 1983, pp. 296–319.
- [5] J. C. C. McKinsey, A. C. Sugar, P. Suppes, *Axiomatic Foundations of Classical Particle Mechanics*, J. Rational Mech. Anal. **2** (1953) 253–272. DOI: 10.1512/iumj.1953.2.52012.
- [6] P. Suppes, *Introduction to Logic*, D. Van Nostrand, Princeton, 1957.
- [7] A. S. Sant’Anna, *The definability of physical concepts*, preprint, PhilSci-Archive item 1967 (2004).
- [8] A. S. Sant’Anna, O. Bueno, *Sets and Functions in Theoretical Physics*, Erkenntnis **79** (2014) 257–281. DOI: 10.1007/s10670-013-9491-y.
- [9] N. C. A. da Costa, A. S. Sant’Anna, *The mathematical role of time and spacetime in classical physics*, Found. Phys. Lett. **14** (2001) 553–563.
- [10] D. J. BenDaniel, *Definability in Physics*, arXiv:physics/0206011 (submitted 2002; final version 2009).
- [11] T. C. Wallstrom, *Inequivalence between the Schrödinger equation and the Madelung hydrodynamic equations*, Phys. Rev. A **49**(3) (1994) 1613–1617. DOI: 10.1103/PhysRevA.49.1613.
- [12] J. P. Ralston, *Quantum Theory without Planck’s Constant*, arXiv:1203.5557 (2012).
- [13] M. Faizal, A. Shabir, A. K. Khan, *Implications of Tarski’s Undefinability Theorem on the Theory of Everything*, EPL (Europhysics Letters) **148** (2024) no. 3, 39001. DOI: 10.1209/0295-5075/ad80c2; arXiv:2410.10903 [physics.hist-ph].